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Environmentally conscious but helpless: Consumer challenges for participation in Formal e-waste collection initiatives of India

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ABSTRACT

India's new e-waste management rules (EWMR) with provisions for extended producer responsibility (EPR) aims to promote formal recyclers, yet dominance of informal sector for recycling waste electrical and electronic equipment (WEEE) continues. Consumer inertia to adopt formal e-waste collection (FEWC) channels is a major reason for this continued dominance. This study explores consumer e-waste disposal behaviours and challenges to participate in FEWC channels through an exploratory research design. Consumer behaviour determinants and barriers are categorized under the framework of the Theory of planned behaviour (TPB) to report variations in e-waste disposal behaviours and participation in FEWC. The study reveal that people are unable to participate in FEWC initiatives despite being environmentally conscious. Poor awareness of FEWC channels and lack of transparency in recycling processes are some crucial pivots that prevent adoption of FEWC channels. The study proposes a conceptual framework for pro-environmental e-waste disposal behaviours and successful implementation of EWMR.

1. Introduction

Literature has for long reported dominance of the informal sector in e-waste recycling, and the ineffectiveness of formal e-waste collection (FEWC) channels and recyclers globally. Lack of consumer participation in FEWC and recycling initiatives is cited as a major reason for this dominance (Anderson and Lee, 2025). Developed nations like U.S., Canada, West Europe, Australia, Japan and South Korea too illegally export their waste electrical and electronic equipment (WEEE) produced through informal channels to developing nations for processing despite having stricter e-waste recycling laws within their national boundaries (Saxena et al., 2025; Forti et al., 2020; Yoshida and Yoshida, 2014; Petridis et al., 2020; Shittu et al., 2021). Dominant and attractive informal e-waste recycling sector (due to lack of tax compliances) force authorised recyclers to function at underutilised recycling capacity and high cost, thereby questioning their survival across the world. The developing nations face the issue of informal recyclers dominance with even more severity as the world's major WEEE is exported to these nations for processing in addition to their own fast paced growth of WEEE (Newaz and Appolloni, 2024; Dutta and Goel, 2021). In the absence of strong regulations, and the dominance of informal recyclers, the developing nations process WEEE in crudest manner leading to dumping of hazardous chemical in dump sites or landfills (Jampala and Shivnani, 2024). To address this grave situation, the Indian government amended the e-waste management rules (EWMR) in 2023 and included provisions for a robust extended producer responsibility (EPR) to encourage formal recycling of WEEE. EWMR aims to end this dominance of informal recyclers by promoting FEWC channels, aggregators, dismantlers and recyclers. The status on successful implementation of these rules and public support and opinion on EWMR and policies though remains to be assessed.

Aggressive marketing to generate demand for electrical and electronic appliances (EEE) along with planned obsolescence of EEE rapidly led to increase of e-waste. The generation of e-waste in India doubled between 2018 and 2022 (Sun, 2023), yet its management is in the nascent stages in the country (Koshta et al., 2022, 2021; Sharma et al., 2024). The quantity of e-waste generated in India was 16,01,155.36 tons during 2021–22, of which only 33 % (527,131.57 metric tons) was collected and processed by the formal sector (MoEFCC, 2023). According to ICEA (2023), the share of WEEE collected and recycled by the formal sector (authorized aggregators, dismantlers and recyclers) stands at just 10 %. An increase in consumption and obsolescence rates of electronic products (Dwivedy et al., 2015) accompanied by low FEWC and recycling (Arya and Kumar, 2020; Dutta and Goel, 2021) is

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broadening the gap between e-waste processing and careless disposal (Forti et al., 2020; Koshta et al., 2021). While India introduced legislative measures, such as the E-waste management policy (2016) (GOI, 2016), to promote the formal collection and recycling of e-waste, the informal sector still dominates the e-waste landscape in the country (Verma et al., 2025; GOI, 2021). The dominance of this informal sector represented by unauthorized dismantlers and recyclers is hard to overcome since they do not follow tax compliances and are thereby able to offer attractive prices to '*kabaadiwalas*' (rag pickers) and informal aggregators (ICEA, 2023). Consequently, the dominance of informal collectors and recyclers of e-waste is noted as a global issue (Borthakur and Govind, 2019; Dutta and Goel, 2021; Faibil et al., 2023), and continues even after the constant legislation and policy measures for proper e-waste management. In India, the EWMR was first introduced in 2016 and has been amended to shift e-waste recycling from informal to formal recyclers sector and ensure disposal of e-waste in an environmentally responsible manner, with robust, broader (including more EEE like solar panels) and stricter compliance measures (MoEFCC, 2023). This amended rule, which came into effect on April 1, 2023, were put in place for responsible e-waste management along with an effective structure for extended producer responsibility (EPR). The amendments in EWMR aims to fulfil the Sustainable Development Goals (SDG) 2030 and calls for extensive responsibilities on producers to create e-waste channels for registered recyclers, procure EPR certificate from registered recyclers and create suitable awareness campaigns. However, the amended EWMR lacks detail mechanism to enforce EPR certifications on producers. Gupt and Sahay (2019) assessed the government policies on waste disposal over the last two decades and concluded that while the 2016 EWMR includes the provisions for EPR they completely ignore the strong and established network of the informal sector that recycles the bulk of WEEE. Informal sector provides employment to a massive population but practice unscientific methods of recycling, causing serious environmental issues (Islam et al., 2025). Further, in most developing economies the original manufacturing companies do not exist, and they export and market their finished electronic products through retailers. Implementation of EPR policies and routing of e-waste to FEWC thus rests on the retail marketing firms or on Producers' responsibility organizations (PRO) and not essentially on original manufacturers (Garlapati, 2016; Faibil et al., 2023). Such problematic issues exist in most developing countries for e-waste management (Bakhiyi et al., 2018; Chandra et al., 2023). Few studies record consumer response to audit existence or effectiveness of e-waste disposal schemes and channels by retail marketing firms as part of EPR initiatives.

To tackle the safe disposal of e-waste, e-waste bins have also been installed by government, since 2016. These bins also provide the consumers with the contact information of the FEWC collectors (e.g. authorized dismantlers) (Fig. 1).

Despite the installation of e-waste bins by the government, consumer participation in formal e-waste disposal has been low (Thukral et al., 2023). Whether this lack of participation is due to poor awareness about these initiatives and absence of environmental consciousness or whether the implementation of government policies and frameworks is yet to achieve a broader practical existence is still not clear. Literature suggests existence of determinants or barriers that impact consumer participation in formal e-waste disposal programs. Determinants are those factors that may motivate an individual towards formal e-waste collection practices (Borthakur and Govind, 2019; Thukral et al., 2023). These are consumer activation factors which may span from attitudinal and cultural norms to economic incentives to break the inertia of people for disposal of e-waste responsibly. From a consumer standpoint, barriers are factors that prevent or inhibit an individual from participating in FEWC practices (Arya and Kumar, 2020; Kumar and Dixit, 2018; Trivedi et al., 2022). Barriers may interject the disposal behaviour journey of a consumer with personal issues and constraints before he reaches the infrastructural doorstep of FEWC channels. However, studies are deficient to comment on how these determinants or barriers interact with consumer cognition to govern behaviours. Studies acknowledge specific determinants and barriers for success of EWMR (e.g. Borthakur and Govind, 2019; Borthakur and Singh, 2021; Koshta et al., 2022; Kumar and Dixit, 2018) but fail to elaborate on the touchpoints and cognitive processes necessary to re-route the e-waste disposal journey of consumers from informal collectors to FEWC channels. EPR provisions include planning of product obsolescence and routing of bulk e-waste to registered recyclers, right at the inception of product design but its success still depends on responsible e-waste disposal behaviour by consumers through FEWC channels. Literature terms such behaviours as 'responsible' or 'sustainable consumption' and 'extended consumer responsibility' (ECR) (e.g. Luchs et al., 2015; Lenzen et al., 2007; Sheu and Choi, 2019). However, ECR



(a) E-waste collection bin at Dehradun, Uttarakhand
Source: Picture clicked by author

(b) E-waste collection drive at National Capital Region
Source: Namo e-waste

Fig. 1. E-waste bins installed near respondents' locality.

concept has so far not been applied to address the irresponsible e-waste disposal behaviours of consumers.

This article postulates that the government EWMR and EPR policies can be effective only if the consumers support those policies and rules (as well as demand suitable provision and infrastructure from government and businesses if its absent) as part of ECR. Collection of e-waste domestically through *'kabaadiwalas'* is the first step that results in the dominance of the informal sector. Thereafter dominos effect takes charge, and e-waste moves on from informal scrap dealers to a pre-established network of informal aggregators, dismantlers and recyclers and authorised recyclers never get a chance to divert the e-waste to their recycling plants. This study thus has two specific objectives. Firstly, to study consumer behaviour determinants and barriers as a pre-requisite for ECR and participation in FEWC initiatives. Secondly, to examine the mediating role of government and businesses to successfully induce ECR for pro-environmental e-waste disposal behaviours in consumers. By doing so the study aims to understand the inertia of consumers towards pro-environmental behaviours and help curtail the dominance of informal sector in e-waste collection and recycling.

2. Theoretical framework

Theories acknowledging environmental consciousness and awareness have significantly contributed to understanding of pro-environmental behaviours. For instance, environmental consciousness and awareness are critical components in the frameworks suggested by Valance Theory (VT) (Peter et al., 1975) and Norm Activation Theory (NAT) (Schwartz, 1977) that are significant to understand pro-environmental e-waste disposal behaviours. Reinforcement of environmental consciousness and awareness of e-waste hazards will motivate a consumer to think "cognitively rational" according to the VT framework. This reinforcement becomes an important determinant after weighing the positive and negative effects of a decision to participate in pro-environmental initiatives. NAT hinges too on attributing responsibility to personal actions. Behavioural Reasoning Theory (BRT), proposed by Westaby (2005), connects the justification of individual decisions based on the negative or positive outcome of a behaviour. However, placid consumer response to EWMR and EPR policies incite certain questions. For instance, does moral responsibility towards environment and proper disposal of e-waste usher consumers towards pro-environmental behaviours or ECR? A closer introspection of behavioural components like attitude and cultural norms is needed to study activation of environmental consciousness into responsible behaviours and ECR. Theory of planned behaviour (TPB) (Ajzen, 1985), a modified version of the theory of reasoned action (TRA) (Fishbein and Ajzen, 1975), lists the necessary psychosocial elements of attitude, subjective norms and perceived behavioural control. TPB has been widely acknowledged by scholars to study waste management behaviours (e.g. Kumar, 2019; Wang et al. 2019; Dixit and Badgaiyan, 2016; Koshta et al., 2022). TPB emphasizes the role of attitude (ATT) (governed by the level of awareness and environmental consciousness), subjective norms (SN) (through cultural manifestations of peers and relations) and perceived behavioural control (PBC) (the ability and belief to achieve completion of a task). All these components of TPB are apt to conceptualize a framework for pro-environmental e-waste disposal behaviour of people. However, while past studies have used TPB and empirically connected consumer intentions with ATT, SN and PBC they are limited to understand triggering nodal points in consumer cognition to adopt or reject pro-environmental behaviours. With changing sociocultural dynamics an introspection of ATT, SN, and PBC norms would help reveal trigger determinants and barriers that may not have been existing earlier nor tested empirically. An in-depth analysis of consumer cognition on evolving policy interventions provides scope to audit government and business initiatives through amended EWMR and EPR regulations. Analysis of consumer cognition helps this study to propose a theoretical extension of TPB that includes government and business initiatives as mediating variables to help redress the poor engagement of consumers with FEWC channels and help implement EWMR.

3. Methodology

Qualitative method to explore the diffidence or participation challenges of consumers towards FEWC programs was considered apt for this study. Most of the empirical studies test hypothesis and model fit on existing factors already identified in literature for responsible e-waste disposal practices. Fresh consumer perspectives would help change the status quo of the dominant informal sector in e-waste recycling. A qualitative enquiry helps reveal the cognitive processes of consumers when they encounter stimuli for determinants or barriers towards pro-environmental e-waste disposal behaviour. This study encapsulates the ATT, SN and PBC norms of consumers in context of e-waste disposal through personal interviews. The authors identified 32 participants, using purposive sampling to ensure a representation from different Indian cities (Tier1, 2, and 3). The selected participants were interviewed during April to July 2024 and were asked questions pertaining to their e-waste disposal behaviour through a semi-structured questionnaire.

A list of electronic gadgets and electrical equipment provided in Schedule I of the Gazette of India (GOI, 2022) was referred to frame the semi-structured interview (Table 1). Various aspects of e-waste, including understanding its categories, environmental and health hazards, disposal methods, storage behaviours, product exchange programs, financial constraints, support from peers, and awareness of government and business initiatives were covered. Each interview lasted for about 40–45 min.

All the interviews were audio recorded after due approval from the participants. Recorded data was converted into transcripts, and the data was cleaned to remove repetitive words, errors, gaps, and pauses. Data was coded using Atlas.ti v24. A thematic analysis was conducted using a hybrid approach that combined both inductive and deductive coding strategies (Fereday and Muir-Cochrane, 2006). Initially, codes were generated directly from the data (inductive), and subsequently, these themes were organized within the theoretical framework of the TPB (deductive). In the first stage, open coding was conducted through close reading of the interview transcripts to identify meaningful data segments. Preliminary codes were assigned without predefined categories, allowing patterns to emerge inductively from the data. In the second stage, initial codes were analyzed for patterns and grouped into related categories. This axial coding enabled the development of broader themes, capturing underlying patterns in participants' perspectives. In

Table 1
Categories of electrical and electronic equipment referred for questionnaire (Source: GOI, 2022).

Information technology and telecommunication equipment	Large and small electrical and electronic equipment
Laptop/ Tablets/ I-pad/ Notebook/ Notepad/ Personal Computers	Dish Washing machines
Electronic data storage devices	Electric cookers/ Stoves/ Grinders/ Kettle
Printers including cartridges	Microwaves, Microwave Oven
Telephones/ Cellular telephones	Electric heating appliances
Routers/ Modems	Electric fanning, exhaust ventilation and conditioning equipment
Inverters/ UPS	Appliances for ironing, mangling and other care of clothing
	Vacuum cleaners
	Hair dryer/ Electric Shaver
Consumer Electrical and Electronics	Toys, Leisure and Sports Equipment
Television sets/ Set-top Boxes	Electrical trains or car racing sets
Refrigerators	Video games/ Hand-held video game consoles
Washing machine	Sports equipment with electric or electronic components
Air- Conditioners	
Digital Cameras/ Recorders	
Audio Amplifiers	

the third stage, these emergent themes were further classified into two overarching categories: determinants and barriers for consumer participation in formal e-waste disposal practices. This categorization was guided by the thematic content as well as the research objective of identifying both facilitating and inhibiting factors. In the final stage, the refined themes were deductively mapped onto the constructs of the TPB (ATT, SN, and PBC). This integration of inductive insights within a theoretical framework enabled a theory-driven interpretation and informed the development of the conceptual model. To ensure the validity of the identified themes and their classification as determinants and barriers, expert validation was conducted prior to mapping them onto the constructs of the TPB. The coding process was subjected to expert validation (Nowell et al., 2017). Four faculty members from Marketing department and department of environmental sciences from author's affiliated institution reviewed the themes for conceptual clarity, consistency with the data, and theoretical alignment. The creation of memos and network maps enabled the comprehensibility of the interplay between the ATT, SN, and PBC norms.

4. Results and findings

Determinants and barriers for engagement and participation in FEWC programs as part of ECR initiatives are listed in this section (Table 2) under the components of TPB framework (ATT, SN, PBC).

4.1. Attitude (ATT)

Participants from Tier 1,2 and 3 cities had a sound understanding of hazards associated with irresponsible disposal of e-waste. They expressed concerns regarding improper disposal of e-waste. One of the participants could relate the contamination of soil along with crops and vegetables, with chemicals from e-waste and consequent rise in cases of diseases like cancer. This environmental consciousness was a trigger determinant to participate in FEWC but was not supported sufficiently by consumer awareness regarding FEWC centres. What should be done to address such environmental concerns was a common dilemma of all participants. This dilemma is seen to be a major barrier for people to move up further to the next level in the responsible e-waste disposal journey. *"First of all, tell me where are the formal channels for e-waste disposal?"* a tier 3 city participant questioned. A tier 2 participant reiterated that government should first establish a system for e-waste disposal and then back it up with awareness campaigns and that campaigns without such a system would be of no use. Another tier 2 participant (retired personnel), said that formal channels for e-waste disposal should reach them rather than expecting people to search for FEWC channels. Participants also reported having seen general campaigns for keeping the country clean (e.g. *Swachh Bharat* campaign) or drives for 'no single-use plastic' but never saw any campaign for responsible e-waste disposal and suggested awareness campaigns like *'Swachh Bharat Abhiyan'* (Clean India campaign) with film stars as ambassadors for such a cause. Many participants favoured government rules to put fines or incentivizing to motivate participation in FEWC initiatives. Some, however felt that accessibility of infrastructure was enough and there should not be any need for incentives or fines for a cause like this.

Participants considered information about the track of e-waste channels, authorized recyclers and their contribution to the economy as crucial to convince them to use the FEWC channels. One of the respondents questioned the existing model of waste collection practices of the government and wondered about the track of e-waste and whether it is actually recycled. Participants from tier 1 cities reported that the municipality vehicle that collects waste had waste segregation compartments (biodegradable and non-biodegradable) in the vehicle itself. Tier 2 and 3 city participants were inconsistent in reporting such segregation in the waste collecting vehicle in their cities. A tier 3 participant said people throw their e-waste in municipal vehicles that

Table 2
Consumer barriers and determinants for Pro-environmental e-waste disposal.

Constructs	Barriers	Determinants
Attitude	Storage of e-waste at homes: Psychological and emotional factors, including confusion, sentimental attachment, and hesitation, that lead to prolonged e-waste storage at home.	Diseases & Health Hazards: Consumer awareness of health risks, chronic disease implications, and hazard management guidelines related to e-waste exposure.
	Habitual Disposal Practices: Ingrained disposal habits (e.g. disposal through '*kabaadiwalas*') marked by inertia, low motivation, and routine practices like mixing e-waste with household waste or open burning	Environmental Consciousness & Concerns: Consumer awareness of environmental damage, pollution impacts, and a sense of responsibility toward ecological preservation.
	Lack of Knowledge & Awareness: Insufficient knowledge and awareness regarding proper e-waste disposal methods and available formal channels, lack of motivation to procure information about FEWC as part of ECR.	Transparency & Traceability: Consumer demand for clear information and processes that are traceable for the handling and disposal of e-waste.
	Privacy & Security Concerns: Fears related to data theft and personal information misuse that discourage consumers from disposing of e-waste through formal channels.	Awareness Programs: Repeated awareness campaigns on e-waste disposal practices, drives, electronics lifecycle education, and public information systems.
		Economic & Non-economic Motivators: Both financial rewards and personal motivations, such as guilt avoidance and satisfaction, that encourage participation in FEWC.
Subjective Norms	Lack of Community Initiatives: Absence of community-based programs in residential areas, limiting collective action and awareness for formal e-waste disposal	Peer Influence: Impact of social circles and peer behaviors on individual decisions related to e-waste disposal.
	Lack of Media Coverage: Insufficient media attention and lack of influential promotion, such as celebrity endorsements or news coverage, for e-waste awareness.	Social Media Influence: Role of social media in shaping e-waste disposal behavior through social approval, recognition, and visibility of responsible actions.
	Deprived Neighborhood Practices: Negative influence of prevalent informal or careless e-waste disposal habits observed within local neighborhoods.	Social validation: Use of social media as a strategic tool for amplifying the desire to be perceived as environmentally conscious and triggering ECR as a social movement.
	Prioritization: Low cognitive priority and discussion on sustainability, environment and pollution.	School & Youth engagement initiatives: Inclusion of pro-environmental awareness and practices in school curriculum, initiatives targeting schools and youth to foster early awareness and active participation in responsible e-waste management practices.
	Lack of Support from Family: Absence of encouragement or participation from family members, which can hinder individual efforts toward responsible e-waste disposal, males less sensitive on responsible e-waste disposal behaviors than females.	
	Lack of Social Learning: Absence of dialogue and shared learning about e-waste within social circles.	

(*continued on next page*)

Table 2 (*continued*)

Constructs	Barriers	Determinants
Perceived Behavioral Control	Lack of Knowledge & Awareness of FEWC channels: Limited awareness about the existence, locations, and functioning of formal e-waste disposal channels and facilities.	Convenience & Accessibility: Convenient, easily accessible disposal options, including doorstep pick-up services, in encouraging formal e-waste participation.
	Trust in Informal Channel: Reliance on and confidence in informal e-waste handlers due to familiarity, convenience, perceived efficiency, perceived reliability, past positive experiences, and attractive incentives.	Information Dissemination: Clear and accessible information to guide consumers on proper e-waste disposal practices and available FEWC channels.
	Municipality e-waste collection barriers: Barriers in municipal systems such as lack of dedicated support, waste stream mixing, and absence of separate e-waste compartments.	
	Disposal Barriers for Elderly: Age-related challenges such as physical limitations, digital illiteracy, and lack of assistance that hinder elderly individuals from engaging in formal e-waste disposal.	
	Lack of Disposal Support and collection by FEWC channels: Role of FEWC channels in providing disposal support and accessible collection points to facilitate responsible e-waste management.	
	Time Constraints & Transportation Cost: Deterrent effect of time investment, inconvenience to travel a distance and transportation costs involved in accessing formal e-waste disposal options.	
Government Initiatives	Lack of Infrastructure: Lack of accessible e-waste collection points, limited local drop-off centers, and inadequate waste segregation systems that hampers responsible e-waste disposal.	Financial Incentives: Financial incentives such as subsidies, rebates, and discounts for consumers, along with funding for formal recyclers and collection agencies to promote responsible e-waste disposal.
	Poor implementation of EPR: Low volumes of e-waste routed from producers or representators of producers to certified recyclers, unclear guidelines and policies for retail marketing companies on EPR.	Fines & Penalties: Imposition of heavy fines and penalties on individuals that fail to comply with e-waste disposal regulations, aim to enforce responsible practices.
	Lack of Government Awareness Campaigns: Absence of government-led awareness campaigns that educate the public on the importance and methods of responsible e-waste disposal.	Monitoring: Monitoring defaulters through installation of cameras near dump sites and landfills, uploading of evidence by citizens on government supported mobile apps, follow up on defaulting individuals or agencies with strong punitive actions.
	Lack of Clear Disposal Guidelines: Absence of clear, standardized guidelines for consumers on how to properly dispose of e-waste, leading to confusion and improper disposal.	
	Mistrust & Skepticism: Consumer skepticism about the effectiveness and transparency of e-waste collection systems, concerns over data security, and mistrust in government efforts to manage e-waste disposal.	
	Lack of strict guidelines and poor policy implementation: Absence of stringent regulations, lack of fine implementation on defaulters, low or no punishment for defaulters and ineffective policy enforcement, which undermines efforts to ensure responsible e-waste disposal.	
	Lack of Government Support: Insufficient government action, limited funding, and lack of visibility in promoting e-waste disposal initiatives, leading to reduced community engagement.	
Business Initiatives	Lack of Awareness of Business Initiatives: Insufficient awareness on EEE disposal, poor promotion of business-led EEE exchange programs aimed at encouraging responsible e-waste disposal, limited consumer engagement.	Exchange Programs: Lucrative exchange programs for WEEE such as trade-ins or discounts, to encourage consumers to dispose of e-waste responsibly.
	Participation in EPR Regulations: Poor implementation of EPR guidelines through retail marketing companies and outsourced agencies or promoters, low volumes of routed through these agents to certified recyclers.	Take back: Business programs that offer take-back services, where consumers can return used or outdated electronic products for responsible disposal or recycling.
	Low incentives: Low financial or no financial incentives offered by businesses to encourage consumer participation in responsible e-waste disposal or EEE exchange programs.	Customer Support: Role of reliable customer support in guiding and assisting consumers through the process of take-back programs.
		Logistical Support: provision of logistical assistance such as transportation, pick-up services, and accessible drop-off points to facilitate smooth e-waste disposal for consumers.

collect all kinds of waste, and they did not know the fate of the e-waste. This doubt gets vetted by another respondent who happened to know the track of the municipality waste collecting vehicle and said that employees dealing with waste do not know anything about e-waste hazards and even if the municipal vehicle has compartments to segregate different types of waste it all gets mixed up at Shishambara, a dumping site in Dehradun. Such observations caused distrust in the waste collection channels and initiatives of the government and justified participants' explanation to give their e-waste to '*kabaadiwalas*', which is convenient and wherein they also get instant financial returns. One of the respondents felt that giving e-waste to '*kabaadiwalas*' was a responsible thing to do as they would either recycle it or make some living out of it and prevent its way on to a dumping site. However, some respondents were not sure about the process and destination of e-waste that is sold to the *'kabaadiwalas'*.

Two behavioural trajectories of the participants were observed after they expressed their inability to participate in FEWC channels (Fig. 2). In the first trajectory, the participants reported having taken the convenient path to informal waste collectors in India (*kabaadiwalas*) to dispose of their e-waste. Disposing of e-waste through *'kabaadiwalas'* was habitual behaviour seen in participants, and some did not even cognitively think much beyond the immediate issue at hand for them, i.e. getting rid of the e-waste. The data strongly highlights this 'habitual' thinking pattern as a major attitudinal barrier along with a lack of awareness of FEWC centres. Further probing of participants revealed the retaining or storage of e-waste, which itself had variant behavioural patterns linked to 'reuse', 'reduce' and 'refuse' behaviour. This is the other behavioural trajectory of participants wherein they did not habitually dispose of their e-waste through '*kabaadiwalas*' but rather stored it in their houses. This e-waste storage behaviour was also seen to have variations across city of residence, gender, age and work profile of participants. Often storage behaviour of participants was linked to consciousness about the environment and sustainability. Such participants were also in a state of confusion regarding the possession and temporary storage of e-waste in their houses. During the interviews, one female participant stated that various electronic items such as headphones, cables, chargers, CD drives, and old laptops had been lying unused at home for a long time. The participant also indicated a lack of knowledge regarding appropriate disposal methods and noted that this pattern of accumulation was common among others in their social circle. One set of participants expressed helplessness to repair their old electronic items. At times, sentimental value also acted as a barrier for respondents to dispose of their e-waste.

Another set of participants reported having passed on the old mobiles or laptops to someone known (generally housemaids or helps) who could not afford a new gadget. Such views indicate 'reuse' or upcycling of electronics and resulting in 'reducing' or 'refusing' to buy new electronic gadgets by the participants (Fig. 2). However, the electronic repair shops and company service centres were not always supportive of such behaviours, nor were they willing to buy/exchange e-waste for a nominal price. Respondents were generally of the view that they get better value for their e-waste through *'kabaadiwalas'* than through exchange of old gadgets for the new ones from the authorised dealers. One tier 2 respondent explained in frustration that going to authorised dealers was a waste of time when '*kabaadiwalas'* give them better deal at their doorstep. Absence of customer support at a granular level from retail marketing firms at this juncture is a lost opportunity to achieve circular economy goals and also evince a gap in implementation of EPR policy (Table 2).

4.2. Subjective norm (SN)

Participation in FEWC by a few friends and relatives of the respondents was noted and appreciated. A female participant from tier 1 city said that her friends and relatives take the collected e-waste in the home and dispose it at designated spots in Connaught Place, Delhi, which inspires her to dispose e-waste in a similar manner. Another tier 2 respondent reported that one of his friends, relocating to a foreign country, had a room full of electronic gadgets for disposal and he handed it directly to an e-waste handing plant in Vizag. One of the participants also talked about posting his picture over social media while disposing of e-waste responsibly and got quite a few likes on his post. Such examples of responsible e-waste disposal behaviour by a friend or relative were seen to motivate the consumer intention to participate in FEWC initiatives immensely. These were a few of the desirable ECR behaviours or intentions reported towards participation in FEWC channels post e-waste generation at household level (Fig. 2). Hence, the practice of using FEWC is not widely propagated as a significant cultural norm. Despite acknowledging irresponsible e-waste disposal as a critical environmental hazard, participants reported that this was not even considered an issue of discussion amongst friends and family. Some participants believed that knowledge about e-waste disposal should be given at the school level.

This study thus notes that while participants ranked high on environmental consciousness the top-of-the mind recall for disposing e-waste through FEWC channels was dormant. Lack of FEWC awareness was often an easy explanation to justify the consumer's diffidence towards FEWC. Poor infrastructure and lack of proper e-waste disposal was seen to dissipate into different types of general complaints from the participants. For instance, a tier 2 participant narrated that on way to his office he daily sees litter dumped by the roadside which has organic waste, wires, batteries, pieces of glass and plastic all mixed up and often animals are seen consuming this dangerous e-waste along with the organic waste. Other respondents also expressed anxiety over such sights and helplessness as there was no place they could report this or do something about it. Some female participants also complained of a lack of support from their family members. A working female expressed her frustration about e-waste stored in the house by saying, "*To me, it does bother, but to my husband, it doesn't affect much*". Time constraints, poor mobility, and the inconvenient location of FEWC added to the list of complaints. A retired housewife said that she is too busy with domestic household responsibilities and that her husband uses most of electronic stuff and also deals with '*kabaadiwalas*' when he has to dispose e-waste or to repair them in some shop. Except for the tier 1 city, the females in

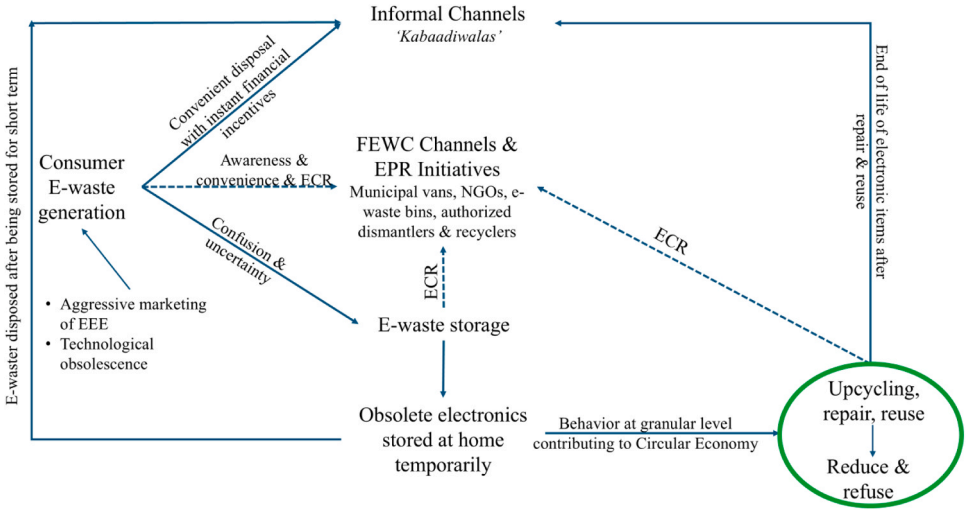


Fig. 2. Consumer e-waste disposal journey.

tier 2 and 3 cities were quite subdued in finding out ways to dispose of their e-waste responsibly. Participants, towards the latter half of the interview, often went on a recommendation track and talked about community initiatives to spread awareness of FEWC in residential colonies and societies or recognizing individuals by the government who participate in FEWC with a certificate like one they got on getting vaccinated for COVID-19. *"If a celebrity or even a political figure campaigns for FEWC like they did for Swachh Bharat Abhiyan, then it would make a difference",* a participant added. According to some participants, the FEWC should be enforced through rules and legal compliance bodies through the imposition of fines and other penalties. In addition to the government initiatives, almost all the participants believed that businesses could play a big role in responsible e-waste disposal. One of the female participants from tier 2 city blamed profit driven businesses for e-waste generation and said that if people don't buy latest gadgets, then they get left out. Retailers and service centres also do not support consumers to repair their old gadgets and express difficulty in getting spare parts. They instead encourage them to buy a new gadget at a very poor exchange value. Almost all participants reflected this sentiment of being forced to buy new electronic gadgets because of technology obsolescence or the service centre's reluctance to repair old gadgets effectively.

4.3. Perceived behavioural control (PBC)

Participant's environmental consciousness about the hazards of improperly disposed e-waste motivated them to segregate e-waste at the source, and a few of them also reported having responsibly disposed of it. One of the respondents reported that he keeps a bag for e-waste at home and when it gets filled, they take it to the office of '*Waste Warrior*' (Waste Warriors, n.d.), which is a non-profit, voluntary citizens' local Non-Governmental Organization (NGO) for waste collection in a tier 3 city. Prior information on the location of the office was crucial as the respondent emphasized that they did not have to plan a special trip to dispose the e-waste but did it whenever they were going in the direction of Waste Warrior office. In tier 1 cities, the e-waste bins in malls or e-waste collection drives on sidewalks (Fig. 1b) (Namo E-waste, 2023) also impacted the perceived ease of disposing of e-waste responsibly. When the availability of infrastructure or centres for e-waste disposal was known to the participants it led to an intention to use those centres in future. *"I have heard about e-waste recycling collection centres that also collect waste from people, but I haven't visited one myself"*, one of the participants explained. Respondents exhibited responsible behaviours wherever the service centres and shopping malls had provisions for e-waste dustbins or repair facilities. A tier 1 city respondent was happy to dispose his old mobile phone in e-waste bin at a retail outlet of Croma though some incentives through such disposals were desirable. However, such reformative retail outlets were largely in tier 1 cities, and the residents of tier 2 and 3 cities did not report the availability of e-waste bins or centres near their locality. According to the participants from tier 2 and 3 cities, smaller electrical shops repair old electronic gadgets and help them in the upcycling of gadgets and 'reuse' behaviours. Some respondents from tier 2 and 3 cities also reported to have repaired their old electronic gadgets in small electronic repair shops. Poor accessibility of the FEWC centres was a major barrier to responsible e-waste disposal, according to the participants who were reluctant to travel a long distance to dispose of their e-waste. "*When we come home in the evening after a long day, we hardly have any time for this task*," a male respondent from tier 2 city expressed their helplessness for his inability to find time to explore formal channels of e-waste collection. For most of the respondent's convenience played a big role and expressed willingness to dispose e-waste through FEWC channels if such channels were located at a nearby place. Residents in tier 1 and tier 2 cities expressed hesitancy to come out of their homes even for pleasurable activities like watching a movie or going to a restaurant due to traffic congestion and crowds. Perhaps the reason why some of the participants suggested financial incentive schemes to offset time constraints and barriers to dispose e-waste through FEWC centres. A female respondent expressed helplessness regarding e-waste and narrated that since they do not have any idea about FEWC channels they pass on their e-waste to someone who can use it or it keeps lying at home until finally it is disposed through '*kabaadiwala*'. Lack of ECR to use FEWC after a electronic gadget's end-of-life (EOL) even after upcycling/reuse/repair is another desirable behavioural change this study notes (Fig. 2). Respondents expected both the government and the businesses to set up e-waste disposal centres for formal collection of e-waste. A possible behaviour to participate in FEWC channels first needs to have a narrative of success in the cognition of people. Without proper awareness of the location of the FEWC centre or the processing or recycling of e-waste the respondents don't feel convinced that their efforts will lead to any positive environmental cause. Consequently, most of the respondents do not seem to even attempt and take serious initiatives towards ECR and conveniently dispose of their waste through '*kabaadiwalas'*. Yet almost all respondents were found to be immensely concerned to dispose of their e-waste through informal channels and had a mixed feeling of guilt and helplessness for not being pro-environmental in their behaviour.

5. Discussion

Helplessness towards responsible disposal of e-waste despite being environmentally conscious, confusion regarding e-waste storage behaviours, desire to re-use, reduce, upcycle EEE, wish to be acknowledged for pro-environmental behaviours, incomplete information on FEWC channels, and behavioural differences due to age and gender, were some of the significant themes that the qualitative data highlights. The impact of irresponsible e-waste disposal on the environment should trigger the "cognitive rational" of participants as suggested in the VT and NAT frameworks and coax them towards ECR. People are expected to be more responsible towards their e-waste disposal practices if they are aware of the consequences of irresponsible e-waste disposal behaviours (Ebreo and Vining, 2001). Interestingly, this study reveals that the stimulus of 'environmental concern' was not seen to be strong enough to transform into a positive attitude and result in consumer intention and responsible behaviour to participate in the FEWC channels. People were seen to be environmentally conscious of the damaging impact of irresponsible e-waste disposal behaviours, yet they are helpless or not motivated enough to dispose e-waste responsibly. Negligible level of awareness about the FEWC channels in their respective cities was one major reason for this. Lack of awareness towards FEWC channels is frequently reported in studies (e.g. Awasthi and Li, 2018; Borthakur and Govind, 2017; Deng et al., 2017; Islam et al., 2021; Thukral et al., 2023) and listed as a primary reason for the failure of FEWC channels. Consequently, people expressed their helplessness or reluctance even to attempt to move away from their usual practice of disposing of e-waste through informal channels ('*kabaadiwalas*') despite being environmentally conscious. Using informal channels for e-waste disposal has always been the usual practice (Borthakur and Govind, 2017; Borthakur and Singh, 2022; Dhir et al., 2021; Pandey et al., 2018) and this study reports similar e-waste disposal behaviour of participants. Participants habitually and culturally agreed to dispose of their e-waste through '*kabaadiwalas*' since it is convenient and gives instant monetary compensation (Fig. 2). Previous studies attribute such behaviours to perpetuating the dominance of informal e-waste recycling channels in India, citing it as a cause for poor acceptance of FEWC (Pandey et al., 2018) and hampering EPR (Dwivedy et al., 2015).

However, lack of awareness of FEWC channels, their inconvenient location or absence makes the informal channels dominant and not the other way round. This was particularly true for tier 2 and 3 cities wherein the FEWC centres were located at far distances. The scarce number of FEWC channels and the inconvenience of reaching out to them prompts consumers to sell their e-waste to informal collectors, leading to mishandling and unsafe disposal practices (de Oliveira Neto et al., 2022; Ikhlayel, 2018). Authorized dismantlers that represent the

FEWC are far too less as compared to the expanse and population of India (Fig. 3). Koshta et al. (2022) indicate that people's PBC over intention to participate in FEWC channels is weak when it is costly, complex and inconvenient to use the FEWC channels. This study too echoes that expecting people to invest time money, and face inconvenience of heavy traffic towards ECR would be a futile hope. Awareness and easy accessibility were seen as crucial factors for using the FEWC channels to dispose of e-waste (Fig. 2). Criticizing irresponsible e-waste disposal behaviours would thus not break the inertia of people to participate in FEWC channels, but providing easily accessible FEWC channels might. This can be a safe assumption as participants acknowledged that informal channels were not the best way to dispose of their e-waste as it finds a way to dumping sites, which is dangerous for the environment, thereby leading them towards guilt and eco-anxiety, a phenomenon that strikes environmentally conscious people with distress due to worsening environmental conditions (Coffey et al., 2021). For instance, waste (including e-waste) gets periodically collected in *Shishambara*, a dumping site in Dehradun and is often incinerated (Tyagi, 2022). Dumping sites and incineration in cities cause eco-anxiety to people and they might opt for a more responsible behaviour to reduce such anxieties. Eco-anxiety through dumping sites and incineration is not directly related to the management of FEWC channels, but the visibility of such sites casts doubts in the consumer chain of thoughts and intentions regarding government policies and the functioning of FEWC channels. People expect transparency on the journey of e-waste, from waste collecting vans to dismantlers and recyclers. Such transparency will strengthen both ATT and SN components and motivate consumers towards FEWC channels (Fig. 4).

Another interesting fact that this study revealed was that while FEWC channels are scarce the ignorance of consumers was also the reason for their lack of awareness about FEWC channels. While participants complained about fewer FEWC centres and channels and the way they function, it was at times observed that many participants were completely unaware of e-waste bins that were installed near their residences. These e-waste bins, though, did not visually appear suitable for collecting e-waste and were seen to have some routine trash (plastic wrappers, etc.) thrown into them instead (Fig. 1a.). Participants do not have a cultural mind-set to think about e-waste and thereby lack the perceptual selectivity to acknowledge pro-environmental initiatives and infrastructure around them and move towards ECR. Previous campaigns like the '*Clean India Campaign*' that encouraged Indians to use public toilets by the Indian government had a similar fate initially as people in rural areas were habituated culturally to defecate in the open (Coffey et al., 2017; Doron and Jeffrey, 2014). Initiatives that prevent environmental degradation are categorized as altruistic causes, which in usual run-of-the-mill activities seldom get prioritized as a stimulus for intention and behaviour. Such initiatives must, therefore, be consistently reinforced in the cognition of consumers through campaigns and policies on FEWC channels. Setting up a number of FEWC channels or transforming informal sector into formal sector for recycling could thus be a pre-requisite but may not be sufficient to make people participate in those channels.

Awareness campaigns in traditional and modern media platforms would reinforce the formal collection of e-waste (e.g. Martinho et al., 2017). Communication messages that inform and remind people on a regular basis of e-waste policies and regulations might activate consumers in their e-waste disposal journey beyond environmental consciousness. While this is the mandate of EPR, under the EWMR rules there is no mechanism to audit whether such campaigns run frequently and widely enough. Not only a check on producer's EPR initiatives would help to raise awareness but government sponsored public service announcements would also push consumers towards ECR (Fig. 4). Celebrities and political figures can feature as social ambassadors in such campaigns to usher consumers towards FEWC channels. Saphores, et al. (2006) attribute significance of demographic characteristics like age, gender, and income to recycling behaviours. Ability, motivation and awareness were seen to vary with age and gender in this study as well. Campaigns thus should not only inform about the hazards of irresponsible e-waste disposal but also facilitate the old, and females to reach out to FEWC channels. Old and retired citizens in households may have limited mobility and alertness but being free from routine employment may be more receptive to such campaigns.

Similarly, female respondents were found to be more knowledgeable and serious in their intention to dispose of e-waste responsibly in this study. Bhatt et al. (2017) also note that women are more active towards responsible e-waste behaviour. Having female workforce in municipality vehicles may thus make females who are alone at home feel more motivated and comfortable to dispose of their e-waste. Campaigns, policies and regulations can articulate recognition, incentives or induce fines for stronger perceptual selectivity and subjective norms (SN) to participate in FEWC programs. People need to be fined severely in case they deviate from norms and regulations. For instance, people are fined just ₹500 (US $ 5) in India for littering, which has an extremely low deterrent impact compared to other countries like Singapore and Japan (Bajpai, 2019). Similarly, a more robust mechanism is needed under EWMR to penalize the producers who fail to acquire EPR certificates from registered recyclers with heavy fines. Incentives may not be only monetary but could also be in the form of recognition certificates like COVID-19 vaccine certificates as mentioned by one of the participants. Campaigns, policies and regulations would emphasize upon ECR for

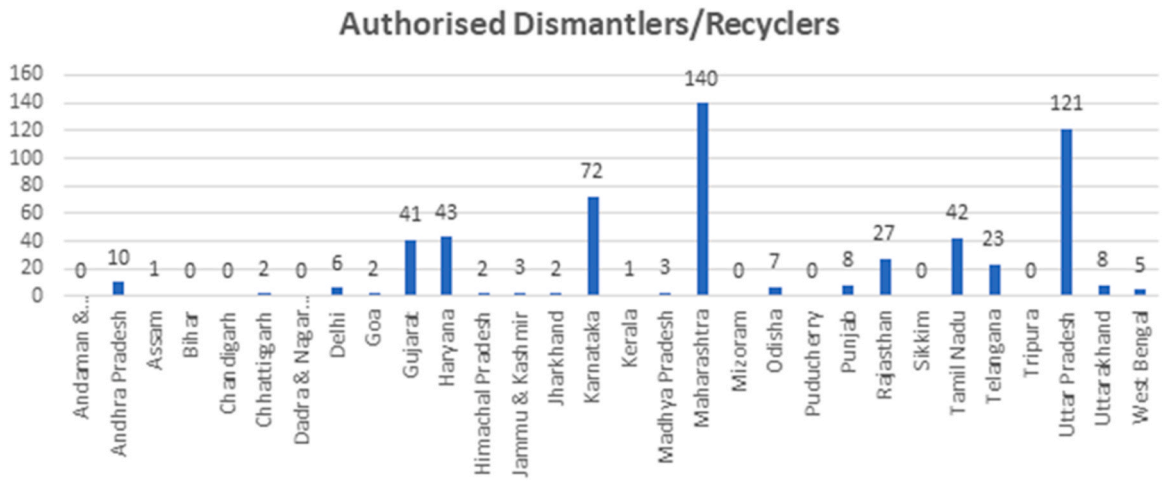


Fig. 3. Number of authorized dismantlers and recyclers of e-waste in India 2023, by state (Source: Sun, 2024).

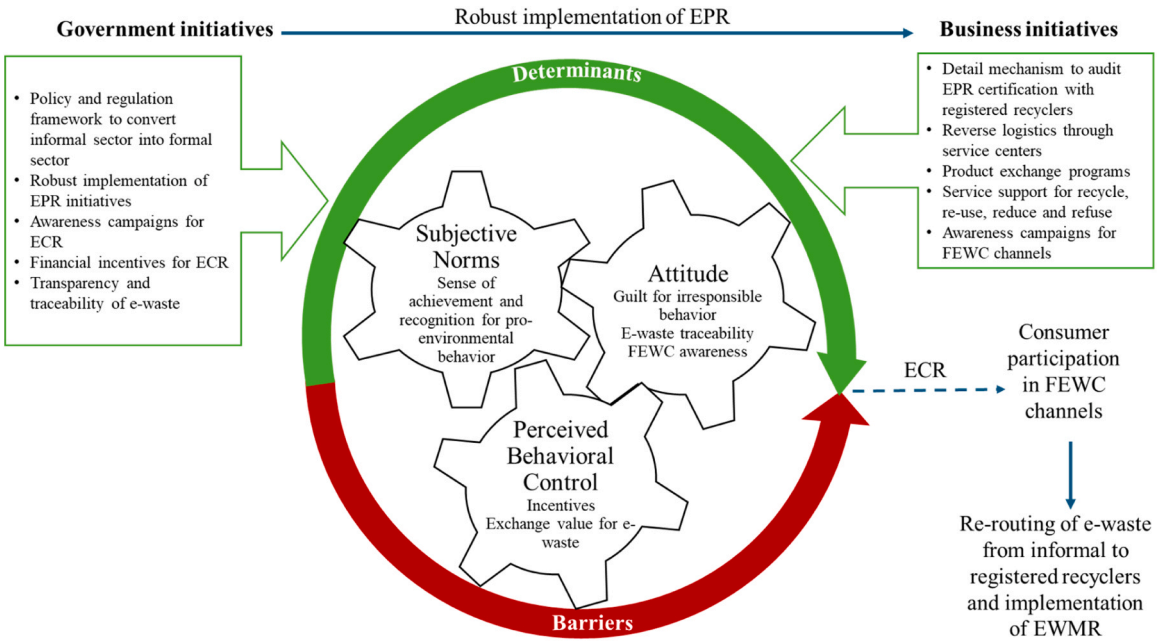


Fig. 4. Proposed framework for successful implementation of EWMR.

e-waste disposal (Sheu and Choi, 2019). Campaigns not only need to address responsible e-waste disposal messages but also inform on waste segregation and processes followed by the dismantlers that prevents leakage of e-waste into landfills. Traceability of e-waste from collection to recycling will strengthen the attitude of consumers to contribute to the efforts of the government. Intention to participate in FEWC will strengthen when people see the implementation of waste segregation happening in waste collection vans, trained personnel to deal with e-waste, an increase in number of suitable e-waste bins (Fig. 1b) and lessening of dumping sites and incidences of incineration in their cities or on highways. E-waste collection drives by private organizations in tier 1 cities (Fig. 1b) (Namo E-waste, 2023) should also happen in tier 2 and 3 cities. This would increase convenience to a larger consumer base and enable them to participate in FEWC channels. Such a change, along with awareness and regulatory enforcement, would make people discuss and create a sort of movement or ripple effect in the subjective norms of society towards pro-environmental behaviour.

When respondents neither disposed of their e-waste through FEWC channels nor informal channels, they were taking a recourse to store the old and obsolete electronic gadgets (Fig. 2). E-waste storage behaviour as reported by numerous studies (e.g. Borthakur and Singh, 2021; Casey et al., 2019; Nowakowski, 2019; Thiébaud et al., 2018; Zhang et al., 2019) is also cited as a barrier for recycling and realization of circular economy goals (Bai et al., 2018; Singh et al., 2018). Nowakowski (2016) points out that storing and reusing is a common behaviour in developing countries. Such storage behaviours were more prominently seen in participants of tier 2 and3 cities. This study also noted repairing or upcycling of the stored e-waste by consumers as a desirable behaviour that leads to the reuse of gadgets. However, retail marketing firms don't appear to encourage such behaviours. Clearly businesses face paradox to either pursue high sales through updated hi-tech electronic gadgets (McManus and Carvalho, 2022) or provide suitable resources for repair and reuse of old gadgets (Sharma et al., 2024). Participants observed that they are forced to buy new gadgets as their old gadgets become obsolete and view this as a selfish linear economy model of the businesses.

The study recognizes that participants are willing to upcycle and repair their old electronic items which leads to 'reuse', 'reduce' and 'refuse', actions towards EEE and which is a pro-environmental intention at a granular level. Long-lasting use of gadgets increases product life cycle, minimizes environmental damage and contributes to circular economy goals (Lyu et al., 2023; Rizos and Bryhn, 2022; Sengupta et al., 2023). Reuse, repair and good exchange value for old gadgets may impact revenues of businesses but this aspect is beyond the scope of this study. However, as an EPR initiative, marketing retail firms can incentivize or make it compulsory for people to return old gadgets before they buy a new one like in Japan (Terazono, 2013). Incentives can offset costs and convenience barriers that hinder PBC over consumer behaviour (Dixit and Badgaiyan, 2016). This will divert consumers from informal channels to FEWC channels in the form of authorized recyclers and dismantlers to dispose of their e-waste. Business-led FEWC channels with suitable incentive schemes and exchange policies can immensely contribute to EPR and circular economy goals (Dixit and Badgaiyan, 2016; Forti et al., 2020; Niti Aayog, 2023; Rizos and Bryhn, 2022) and prevent leakage of e-waste into dumping sites through informal channels (ICEA, 2023). However, success of EPR would be much enhanced with adequate infrastructure and formalising the informal recyclers and dismantlers in the e-waste management (Bakhiyi et al., 2018). While the barriers and determinants for participation in FEWC are discussed here under the TPB framework, these are not to be understood as isolated factors but as enmeshed variables within the ATT, SN and PBC components of TPB (Fig. 4). For instance, while environmental consciousness wasn't enough to trigger a positive attitude towards intention to participate in FEWC channels, the transparency in the journey of e-waste, recognizing positive e-waste behaviours and prevention of dumping sites can impact subjective norms that would impact both attitude and perceived behavioural control. Similarly, awareness of FEWC channels, policies, regulations, fines and incentives would eliminate consumer ignorance, activate the attitude component and enable perceived behavioural control. Thus, the ATT, SN, and PBC components of TPB do not interact sequentially or in a linear fashion but trigger each other as flywheels to break the consumer inertia for participating in FEWC initiatives. The government and business initiatives act as catalysing factors to trigger ATT, SN and PBC components of TPB (Fig. 4).

6. Conclusion

The study reveals that consumer may achieve cognition for pro-environmental behaviours, provided the GI and BI interventions crucially mediate the cognitive variables (ATT, SN, PBC) of TPB. Strict

norms on retail marketing firms for EPR certification, efficient reverse logistics and attractive product exchange schemes, supported by consumer awareness campaigns would strengthen EPR norms. The study suggests the need for strong government initiatives for a robust mechanism to implement EPR guidelines and reinforcement of ECR through fines, recognition, rewards, monetary incentives and awareness campaigns towards FEWC channels. Consistent awareness campaigns over traditional and social media on FEWC centres, by celebrities endorsing e-waste recycling and showing documentary evidence of e-waste recycling would not only engage or remind but also enable consumers towards FEWC channels. Regulatory measures like fines or recognition of compliance towards FEWC through certification can reinforce ECR and motivate consumers to break their inertia and move away from '*kabaadiwalas*' to FEWC channels. Supporting pro-environmental e-waste disposal behaviours of friends and relatives and encouraging behaviour change to participate in FEWC needs to be reinforced by consumers culturally as ECR. Further, ECR does not only include pro-environmental e-waste disposal, but also demand strict implementation of EPR or rectification in EWMR from the government as a responsible citizen. Enforcement of regulatory metrics to track e-waste generated through retail marketing firms or Producers' responsibility organizations (PRO) can make EPR initiatives and EWMR more robust. Producers can introduce policy for installation of e-waste bins at retail centres and reinforce EPR norms of the government.

Integration of qualitative insights with TPB contributed to the theoretical advancement by contextualizing behavioural constructs (ATT, SN, and PBC) for better assessment and prediction of intention towards e-waste disposal behaviours. As future scope new behavioural nuances along with the mediating impact of government and business initiatives on ECR and EPR can be empirically examined for the fitness of the proposed framework. This would enable policy makers and regulatory agencies to administer the scale and suitability of operations needed to manage the FEWC channels and implement EWMR successfully.

CRediT authorship contribution statement

Mithil Kosarkar: Writing – review & editing, Writing – original draft, Visualization, Methodology, Investigation, Formal analysis, Data curation. **Sunil Barthwal:** Writing – review & editing, Writing – original draft, Visualization, Supervision, Resources, Methodology, Investigation, Formal analysis, Conceptualization.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper. The authors declare the following financial interests/personal relationships which may be considered as potential competing interests:

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Data availability

Data will be made available on request.

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